higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING

NQF LEVEL 4

(12010464)

18 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages and 1 diagram sheet.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Rule off across the page after each completed question.
6. Use only blue or black ink.
7. Write neatly and legibly.

Copyright reserved

QUESTION 1: CONSTRUCTION CONTRACTS

1.1 The Construction Industry Development Board prescribes tender procedures that all government institutions must follow during the tendering and contract awarding process.

Explain the correct tendering procedures to be followed during the tendering process only. (7)

1.2 During a construction project, the contractor builds a structure for the client according to the design and specifications of the professional team.

Name FOUR professional consultants who may be involved in the construction project. (4)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (1.3.1–1.3.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.3.1	A person who will help to settle an argument	A	arbitrator
1.3.2	A person who will decide who is correct when a dispute breaks out with respect to the agreement	B	mediator
1.3.3	A person that will help to settle an argument based on the agreement	C	adjudicator
1.3.4	A person who is appointed by the employer to do specialist work on a site	D	contract instructor
		E	direct contractor

(4 × 1) (4)

[15]

QUESTION 2: BASIC COMMON LAWS APPLICABLE TO CONSTRUCTION

2.1 Explain FOUR responsibilities of land owners in terms of excavation with respect to lateral support of neighbouring property. (4)

2.2 The following paragraph refers to a builder's lien. Complete the paragraph by writing down the missing word or words. Write only the word or words next to the question number (2.2.1-2.2.6) in the ANSWER BOOK.

A builder's lien is a (2.2.1) … that comes into effect when the (2.2.2) … takes (2.2.3) … of the (2.2.4) … and starts construction. The builder's lien is usually invoked when the (2.2.5) … runs into financial problems and cannot (2.2.6) … the contactor. (6 × 1) (6)

[10]

QUESTION 3: STANDARD CONSTRUCTION CONTRACTS

3.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

3.1.1 A contractor can decide to use a precast concrete lintel in an opening if there are no specifications for it.

3.1.2 The contractor has the right to be paid for alterations and variations when there are changes to the scope of work of the project.

3.1.3 A nominated subcontractor is selected from a list of contractors supplied by the employer.

3.1.4 It is not important that variation orders are signed.

3.1.5 Contractors keep record of work progress which can be used for extension of time if delays occur.

(5 × 1) (5)

3.2 List FOUR methods that a contractor will use to verify the progress of work on a building project. (4)

3.3 Give the full wording of the abbreviation JBCC as used in the construction industry. (1)

[10]

Copyright reserved

QUESTION 4: TENDERS AND QUOTES

4.1 Give ONE word or term for each of the following descriptions by choosing a word or term from the list below. Write only the word or term next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

restricted tender; dayworks contracts; lump sum contracts; cost plus contracts; labour only contract; open advertised tender

4.1.1 These contracts are often used for small jobs. The employer buys all the material and the contractor only has to provide the labour.

4.1.2 These tenders are open to all qualified bidders and are publically advertised. Except under exceptional circumstances, all government work is tendered this way.

4.1.3 A client from the private sector chooses contractors whom he would like to work on his project and only invites these people to tender.

4.1.4 This is an agreement between parties on the hours of work done by each man and the materials used. The contractor is paid for the labour and materials plus an agreed percentage for overheads and profits.

4.1.5 These tenders are used where there is a lot of uncertainty because the site conditions have not been investigated or because the contract documents are not detailed enough for the tenderer to make a proper submission.

(5 × 1) (5)

4.2 Explain the importance of the *scope of work* that is included in the tender documents. (5)

4.3 State any THREE aspects that will affect a tender price. (3)

4.4 The bill of quantities is a tender document that contractors use for pricing.

Name any TWO columns given on the document (billing sheet) before the quantity surveyor and contractor start to fill in the details. (2)

[15]

QUESTION 5: QUANTITIES AND COSTING

5.1 The NCV level 4 classes want to build a simple classroom for the community school.

Study the specifications for the community school building below and the drawings on the attached DIAGRAM SHEET and answer the questions.

The classroom is built with a 220 mm thick brick wall and has no internal walls. The sizes and number of windows on the south elevation are the same as that on the north elevation. There is no door on the south elevation. The length of the south and north elevations is 8 m on each side. The length of the east and west elevations is 4,32 m on each side. There are no windows or doors on the east and west elevations.

Assume there are 104 bricks in a square meter

5.1.1 Calculate the total number of bricks required to complete the superstructure of the community classroom, including the gable walls.
NOTE: Use the centreline method (14)

NOTE: Deduct ALL window and door openings (6 windows and 1 door).
Use the following window sizes:
4@ 1200 mm wide X 1200 mm high
2@ 1600 mm wide X 1200 mm high

5.1.2 Calculate the total cost to build the superstructure brickwork from below the floor level and including the gable walls if the cost of mortar and labour is R165,00 per square meter and the price per brick is R5,50. (4)

5.2 Calculate the inside perimeter of the classroom. (2)
[20]

QUESTION 6: PLANNING AND ORGANISING CONSTRUCTION ACTIVITIES

6.1 A contractor must arrange ALL construction activities in a logical and progressive sequence.

Read through the list of activities, required to erect a simple single storey building and answer the questions.
NOTE: Activities are not in a logical or progressive sequence.

- Nail down roofing sheets
- Erect temporary toilets
- Set out the building
- Place concrete slabs
- Provide canteen and changing facilities
- Clear the site of all vegetation and boulders
- Fence off the site
- Place concrete foundations
- Build foundation walls
- Start with superstructure
- Provide containers and storage facilities
- Excavate foundation trenches
- Install telephone lines
- Backfill and compact soil within foundation walls

6.1.1 Define *progressive sequencing* as used in construction planning. (2)

6.1.2 List FIVE activities that will be conducted during the site establishment stage. (5)

6.1.3 Select and re-arrange the construction activities that will be done from site clearance to the start of the superstructure in a progressive and logical order. (7)

6.2 Critical and non-critical activities need to be planned for a construction project to run smoothly and be completed within the stipulated time.

6.2.1 Name the type of activities that will affect the end date of the project if it is delayed.

6.2.2 Name the type of activities that can be delayed without affecting the end date of the project.

(2 × 1) (2)

6.3 The following activities can each be classified as one of either critical or non-critical. Name the type of activity in each of the following cases.

6.3.1 Laying the foundations

6.3.2 Landscaping the garden

6.3.3 Installing the damp-proof membrane

6.3.4 Having the roof structure inspected

(4 × 1) (4)
[20]

QUESTION 7: PROJECTIONS AND CASH FLOW

7.1 Name FOUR activities which would normally appear on a bar chart scheduling the activities on a building project. (4)

7.2 ABC Contractors have won a tender for the renovation of a community hall with a timeframe of 5 months. The total cost of the project is R250 000. The contract stipulates that ABC Contractors will receive a progress payment of R45 000 in the first month. Thereafter monthly payments will increase by 15%.

7.2.1 Calculate the payment that ABC contractors will receive at the end of the second month.

7.2.2 Calculate the payment that ABC contractors will receive at the end of the third month.

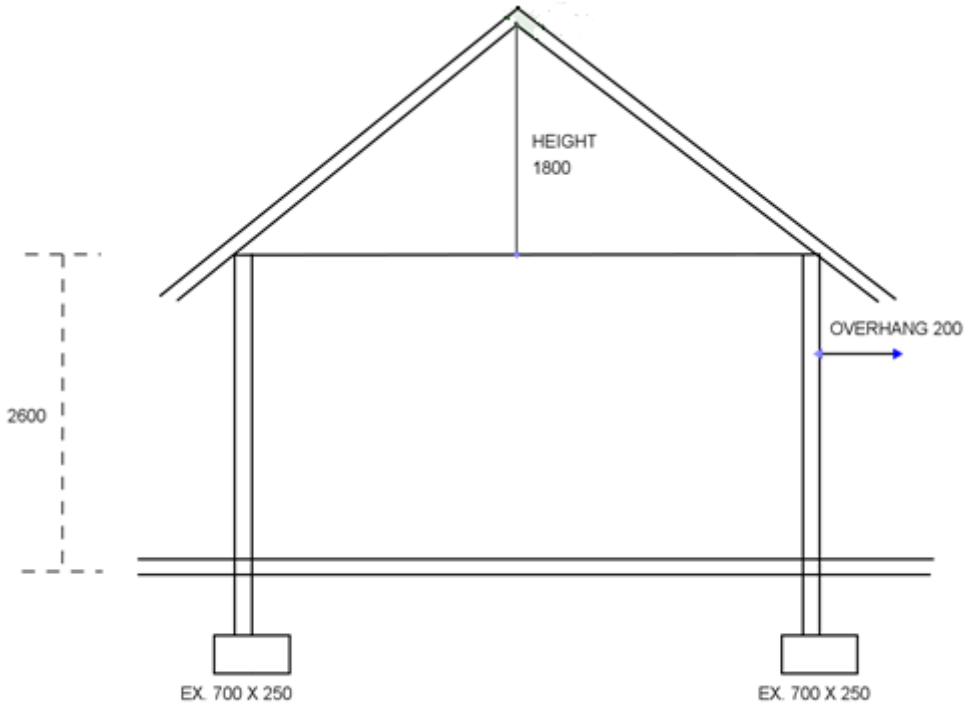
(2 x 3) (6)
[10]

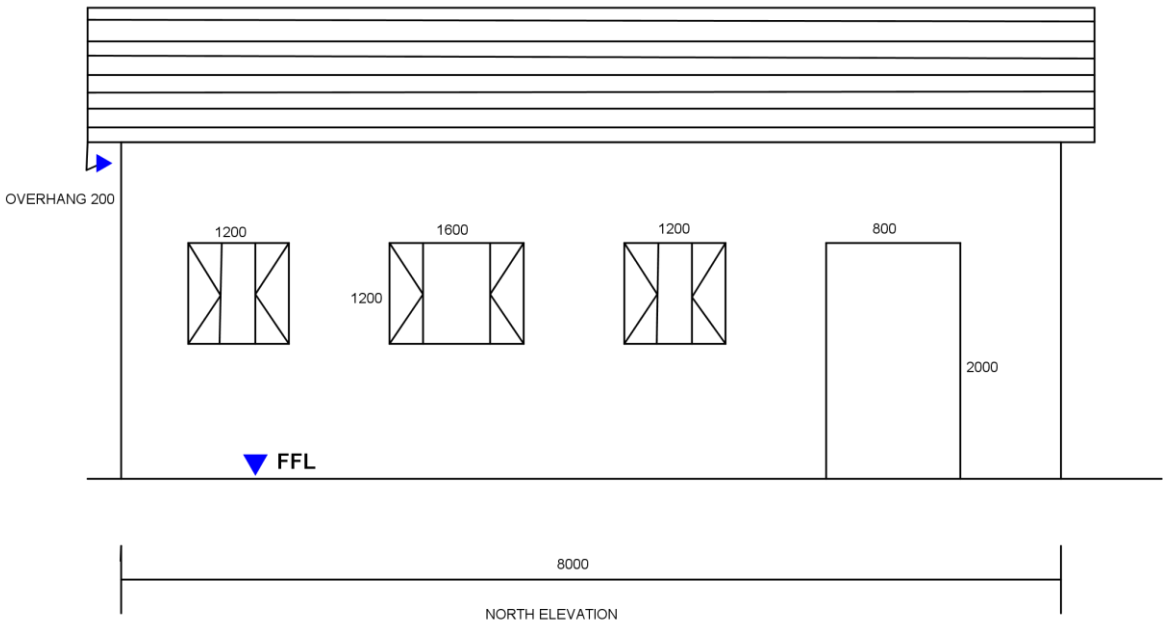
TOTAL: 100

Copyright reserved

(12010464)

DIAGRAM SHEET





Copyright reserved